Parametric Hypothesis Tests — Part 2

The p-value and Further Examples

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3.2 — The p-value

Motivation

Before the break, we used the **critical region** approach:

1. Fix α
2. Find the critical value(s)
3. Compare z_{obs} (or t_{obs}, q_{obs}) with the critical value

But this only tells us "reject" or "do not reject" for a **specific** α.

What if someone asks: *"Would you still reject H_0 at $\alpha = 0.01$?"*

The p**-value** answers this question for **all possible** significance levels at once.

Definition of the p-value

p-value

The p**-value** is the probability, computed under H_0, of observing a test statistic value **as extreme as, or more extreme than**, the value actually observed.

It is the **smallest significance level** at which we would reject H_0.

Informally: the p-value measures *how surprised we should be* by the sample data, if H_0 were true.

- Small p-value $\Rightarrow$ data is very unlikely under H_0 $\Rightarrow$ strong evidence against H_0
- Large p-value $\Rightarrow$ data is consistent with H_0 $\Rightarrow$ no evidence against H_0

p-value Formulas

The formula depends on the type of test:

Type of test	**p-value**
Left-tailed ($H_1 : \theta < \theta_0$)	$p = P(T \leq t_{obs} \mid H_0)$
Right-tailed ($H_1 : \theta > \theta_0$)	$p = P(T \geq t_{obs} \mid H_0)$
Two-tailed ($H_1 : \theta \neq \theta_0$)	$p = 2 \times P(T \geq \lvert t_{obs} \rvert \mid H_0)$

Here T denotes the test statistic (could be Z, T, Q, etc.) and t_{obs} is its observed value.

Decision Rule Using the p-value

p-value decision rule

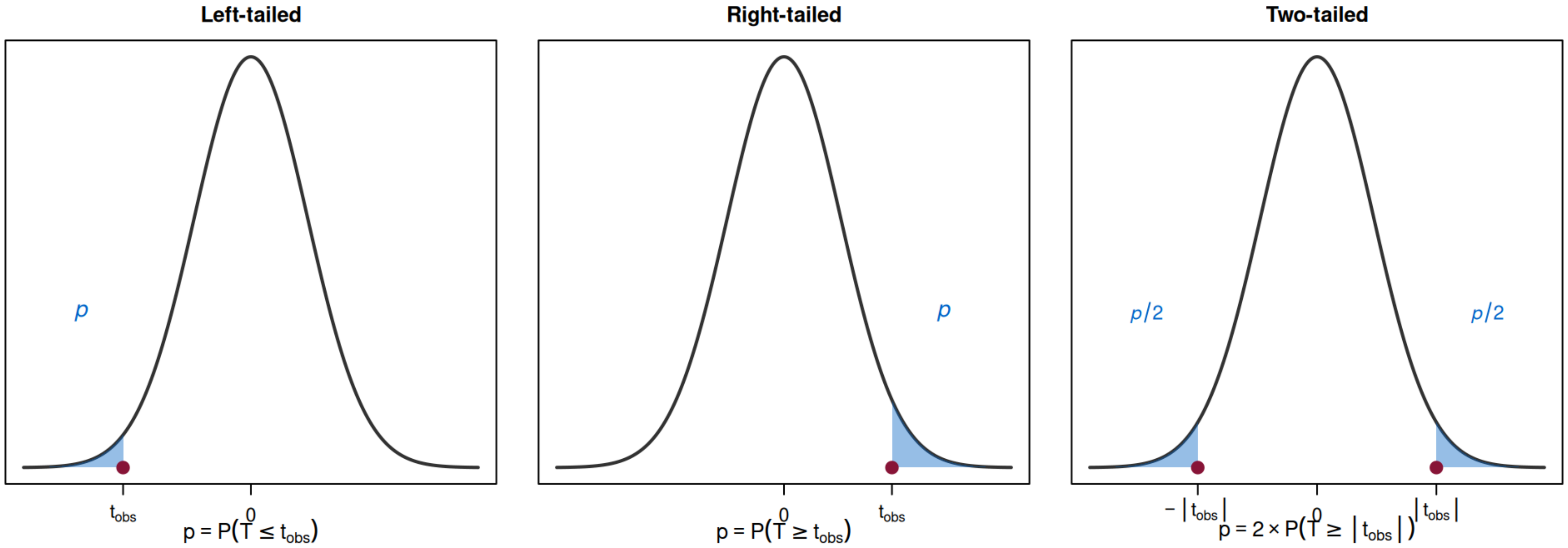
Given a significance level α:

- If $p\text{-value} \leq \alpha \Rightarrow$ **reject** H_0
- If $p\text{-value} > \alpha \Rightarrow$ **do not reject** H_0

This is equivalent to the critical region approach, but more informative:

- The critical region approach gives a binary answer for one specific α
- The p-value tells you **exactly how much evidence** the data provides against H_0

p-value: Graphical Interpretation



The shaded blue areas represent the p-value: the probability of obtaining a result as extreme as t_{obs}, under H_0.

Example 3: p-value Calculation (cont. from Example 1)

Recall: $H_0 : \mu \geq 500$ vs. $H_1 : \mu < 500$, $z_{obs} = -2.25$, left-tailed test.

$$p\text{-value} = P(Z \leq z_{obs} \mid H_0) = P(Z \leq -2.25)$$

$$= \Phi(-2.25) = 1 - \Phi(2.25) = 1 - 0.9878 = 0.0122$$

Interpretation: If H_0 were true ($\mu = 500$), the probability of observing a sample mean as low as (or lower than) 497 is only 1.22%.

Decision (at $\alpha = 0.05$): Since $p = 0.0122 < 0.05 = \alpha$, we reject H_0.

Decision (at $\alpha = 0.01$): Since $p = 0.0122 > 0.01 = \alpha$, we do **not** reject H_0.

Example 4: p-value with the t-distribution

A random sample of $n = 20$ observations is drawn from a Normal population. The sample mean is $\bar{x} = 101$ and the corrected sample standard deviation is $s' = 3$.

Test $H_0 : \mu \leq 100$ vs. $H_1 : \mu > 100$.

Example 4: Solution

Test statistic (σ unknown, Normal population):

$$T_0 = \frac{\bar{X} - \mu_0}{S'/\sqrt{n}} \underset{\text{under } H_0}{\sim} t_{(n-1)} = t_{(19)}$$

Observed value:

$$t_{obs} = \frac{101 - 100}{3/\sqrt{20}} = \frac{1}{3/4.472} = \frac{1}{0.6708} \approx 1.49$$

Example 4: Solution (cont.)

p**-value** (right-tailed test):

$$p = P(T > 1.49 \mid H_0)$$

Using the t-Student table (Table 7) with 19 degrees of freedom:

We look for the row $\nu = 19$. We find that $t_{0.10,(19)} = 1.328$ and $t_{0.05,(19)} = 1.729$.

Since $1.328 < 1.49 < 1.729$, we have:

$$0.05 < p < 0.10$$

Decision (at $\alpha = 0.05$): Since $p > 0.05$, we **do not reject** H_0.

Conclusion: At the 5% level, there is **not sufficient evidence** to

Reading the p-value from Tables

When using the t-Student or χ^2 tables, we typically cannot compute the exact p-value. Instead, we **bracket** it:

Strategy: Find the two table entries that “sandwich” t_{obs} (or q_{obs}).

Example: $t_{obs} = 2.861$, $\nu = 19$ (right-tailed).

From Table 7: $t_{0.005,(19)} = 2.861$.

So $p = 0.005$ (exactly, in this case).

> **Tip**: If t_{obs} matches a table entry exactly, the p-value is the corresponding tail probability. Otherwise, state the interval.

Example 5: Two-tailed p-value

A manufacturer specifies that the mean diameter of a component is $\mu_0 = 25$ mm. The population variance is known: $\sigma^2 = 4$. A random sample of $n = 49$ components has $\bar{x} = 25.6$ mm.

Test $H_0 : \mu = 25$ vs. $H_1 : \mu \neq 25$. Compute the p-value.

Example 5: Solution

Test statistic (σ known):

$$Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$$

Observed value:

$$z_{obs} = \frac{25.6 - 25}{2/\sqrt{49}} = \frac{0.6}{2/7} = \frac{0.6}{0.2857} \approx 2.10$$

p**-value** (two-tailed):

$$p = 2 \times P(Z \geq |z_{obs}|) = 2 \times P(Z \geq 2.10)$$

$$= 2 \times [1 - \Phi(2.10)] = 2 \times (1 - 0.9821) = 2 \times 0.0179 = 0.0358$$

Decision: $p = 0.0358 < 0.05$, so we reject H_0 at $\alpha = 0.05$. There is evidence that $\mu \neq 25$ mm.

Equivalence: Critical Region vs. p-value

Both approaches always give the **same conclusion** for a given α:

Approach	Reject H_0 if…
Critical region	t_{obs} falls in the rejection region
p-value	$p\text{-value} \leq \alpha$

In practice, many practitioners prefer the p-value because:

- It provides **more information** (the exact strength of evidence)
- It allows the reader to draw their own conclusion for their preferred α
- It is the standard output of statistical software

Common Misinterpretations of the p-value

The p-value is NOT:

- The probability that H_0 is true
- The probability that H_1 is false
- The probability of making an error

The p-value IS:

The probability of observing data as extreme as (or more extreme than) what was observed, **assuming H_0 is true**.

Practical Guidelines for Interpreting p-values

While the decision rule ($p \leq \alpha \Rightarrow$ reject) is clear-cut, some authors provide informal guidelines:

p-value	Evidence against H_0
$p > 0.10$	No evidence
$0.05 < p \leq 0.10$	Weak (marginal) evidence
$0.01 < p \leq 0.05$	Moderate evidence
$0.001 < p \leq 0.01$	Strong evidence
$p \leq 0.001$	Very strong evidence

Source: adapted from Newbold, Carlson & Thorne.

> These are informal guidelines, not strict rules. The choice of α remains a decision of the researcher.

Relationship Between Confidence Intervals and Hypothesis Tests

The Connection

There is a direct relationship between a **two-tailed** hypothesis test and a **confidence interval**:

CI–Test Equivalence

At significance level α, we **reject** $H_0 : \mu = \mu_0$ (two-tailed) if and only if μ_0 **does not belong** to the $(1 - \alpha) \times 100\%$ confidence interval for μ.

This makes intuitive sense: if the hypothesized value μ_0 is outside the plausible range given by the CI, then the data contradicts H_0.

Example 6: CI and Test Equivalence

Recall Example 5: $\mu_0 = 25$, $\sigma = 2$, $n = 49$, $\bar{x} = 25.6$.

The 95% confidence interval for μ is:

$$\bar{x} \pm z_{0.025} \cdot \frac{\sigma}{\sqrt{n}} = 25.6 \pm 1.96 \times \frac{2}{7} = 25.6 \pm 0.56$$

$$\text{CI} = [25.04\,;\ 26.16]$$

Since $\mu_0 = 25 \notin [25.04\,;\ 26.16]$, we reject $H_0 : \mu = 25$ at $\alpha = 0.05$.

This is consistent with our earlier finding ($p = 0.0358 < 0.05$).

Solved Exercises

Exercise 1

A manufacturer claims that light bulbs last, on average, at least 1000 hours. A random sample of 25 bulbs from a Normal population gives $\bar{x} = 980$ hours and $s' = 40$ hours.

At $\alpha = 0.05$, is there evidence against the manufacturer's claim?

Exercise 1: Solution

Step 1: $H_0 : \mu \geq 1000$ vs. $H_1 : \mu < 1000$ (left-tailed)

Step 2: $\alpha = 0.05$

Step 3: σ unknown, Normal population, so:

$$T_0 = \frac{\bar{X} - \mu_0}{S' / \sqrt{n}} \underset{\text{under } H_0}{\sim} t_{(24)}$$

Exercise 1: Solution (cont.)

Step 4: Left-tailed, $\alpha = 0.05$, $\nu = 24$

$$\text{Rejection region: } \left]-\infty\,;\, -t_{0.05,(24)}\right] = \left]-\infty\,;\, -1.711\right]$$

(From Table 7: $t_{0.05,(24)} = 1.711$)

Step 5:

$$t_{obs} = \frac{980 - 1000}{40/\sqrt{25}} = \frac{-20}{8} = -2.50$$

Step 6: $t_{obs} = -2.50 < -1.711 \rightarrow$ falls in the rejection region.

Decision: Reject H_0.

Exercise 1: p-value

$$p = P(T \leq -2.50) = P(T \geq 2.50)$$

From Table 7, $\nu = 24$: $t_{0.01,(24)} = 2.492$ and $t_{0.005,(24)} = 2.797$.

Since $2.492 < 2.50 < 2.797$, we have $0.005 < p < 0.01$.

Conclusion: At 5%, there is strong evidence that the mean lifetime is less than 1000 hours ($p < 0.01$).

Exercise 2

A food processing company fills cereal boxes labeled as 500 g. The variance of the filling process is known to be $\sigma^2 = 25 \text{ g}^2$. A random sample of $n = 64$ boxes yields $\bar{x} = 498$ g.

a. Test whether the mean weight differs from 500 g at $\alpha = 0.05$.

b. Compute the p-value.

Exercise 2: Solution

a) $H_0 : \mu = 500$ vs. $H_1 : \mu \neq 500$ (two-tailed)

$\sigma = 5$ is known, $n = 64$:

$$Z_0 = \frac{\bar{X} - 500}{\sigma/\sqrt{n}} \sim N(0,1) \qquad z_{obs} = \frac{498 - 500}{5/\sqrt{64}} = \frac{-2}{5/8} = \frac{-2}{0.625} = -3.20$$

Rejection region (two-tailed, $\alpha = 0.05$):

$$]-\infty;\ -z_{0.025}] \cup [z_{0.025};\ +\infty[\ =\ ]-\infty;\ -1.96] \cup [1.96;\ +\infty[$$

Since $z_{obs} = -3.20 < -1.96$, we **reject** H_0.

Exercise 2: Solution (cont.)

b) p-value (two-tailed):

$$p = 2 \times P(Z \geq |-3.20|) = 2 \times P(Z \geq 3.20)$$

$$= 2 \times [1 - \Phi(3.20)] = 2 \times (1 - 0.9993) = 2 \times 0.0007 = 0.0014$$

Very strong evidence against H_0: the p-value is much smaller than any conventional α.

Exercise 3

From a random sample of $n = 20$ observations drawn from a Normal population, one obtains $\bar{x} = 101$ and $s' = 3$.

Consider the test: $H_0 : \mu \leq 100$ vs. $H_1 : \mu > 100$.

a. For $\alpha = 0.05$, determine the rejection region.

b. If, for another sample of the same size, $t_{obs} = 2.861$, what is the p-value?

Exercise 3: Solution

a) Right-tailed test, $T_0 \sim t_{(19)}$:

$$\text{Rejection region: } \left[t_{0.05,(19)}\,;\,+\infty\right[= [1.729\,;\,+\infty[$$

(Table 7: $t_{0.05,(19)} = 1.729$)

b) For $t_{obs} = 2.861$, right-tailed:

$$p = P(T > 2.861)$$

From Table 7, $\nu = 19$: $t_{0.005,(19)} = 2.861$.

So $p = 0.005$.

There is evidence to reject H_0 for all usual significance levels ($0.005 < \alpha$, for any conventional α).

Summary of Today's Lecture

Key takeaways — Lecture 6

1. A hypothesis test is a formal procedure to evaluate a claim about a population parameter
2. H_0 (status quo) vs. H_1 (research hypothesis); H_0 always contains equality
3. The **test statistic** is a pivot variable computed under H_0
4. **Type I Error** (α) = rejecting true H_0; **Type II Error** (β) = not rejecting false H_0
5. The p**-value** is the smallest α at which we would reject H_0
6. Decision: reject H_0 if $p \leq \alpha$, or equivalently, if t_{obs} falls in the rejection region
7. A two-tailed test at level α and a $(1 - \alpha)$ CI give equivalent conclusions

Next Lecture (April 21)

We will apply the testing framework to specific cases:

- **Section 3.3**: Tests for means and variances of Normal populations
- **Section 3.4**: Tests for the difference of means of two Normal populations
- **Section 3.5**: Large-sample tests (asymptotic normality)

Make sure to review the relevant sections in Newbold (Chapter 9).

Disclaimer

These slides are a free adaptation of the course material for Estatística II by Prof. Teresa Ferreira and Prof. Sandra Custódio from the Lisbon Accounting and Business School — Polytechnic University of Lisbon.

Primary reference: Newbold, P., Carlson, W. & Thorne, B. — *Statistics for Business and Economics*, Global Edition.